A 40-year annual time series of forest type and leaf type cover for Italy derived from Landsat using deep learning

Benedikt Hiebl[1], **Nicola Alessi**[2], **Gianmaria Bonari**[3], **Giacomo Calvia**[4], **Alessandro Bricca**[5], **Giulio Zangari**[4], **Stefan Zerbe**[6,7], **and Martin Rutzinger**[1]

[1]Department of Geography, University of Innsbruck, Innsbruck, Austria
[2]Italian National Institute for Environmental Protection and Research ISPRA, Rome, Italy
[3]Department of Life Sciences, University of Siena, Italy
[4]Faculty of Agricultural, Environmental and Food Sciences, Free University of Bozen-Bolzano, Italy
[5]School of Biosciences and Veterinary Medicine, University of Camerino, Italy
[6]University of Applied Sciences and Arts (HAWK), Faculty of Resource Management, Göttingen, Germany
[7]University of Hildesheim, Institute of Geography, Hildesheim, Germany

Correspondence: Benedikt Hiebl (Benedikt.Hiebl@uibk.ac.at)

Abstract. Consistent long-term maps of forest functional type and dominant tree species cover at high spatial resolution are currently unavailable for Italy. The study presents annual 30 m resolution maps of continous evergeen broad-leaved, deciduous, and coniferous leaf type cover fractions and dominant tree species class probabilities for all forested pixels in Italy spanning 1987–2025, derived from the Landsat Collection-2 Level-2 archive (Landsat 4/5/7/8/9). Two Time Series Transformer models are trained via a three-stage workflow combining contrastive pretraining with supervised fine-tuning on Italian forest plot observations, forest inventories, and a synthetically generated training dataset to predict both tasks. Pixel-level uncertainty estimates are provided as the standard deviation of a five-seed deep ensemble for both models. A three-year observation pooling strategy is applied to increase cloud-free observation density and mitigate inter-annual variability. Evaluation of the time series maps are based on hold-out areas and a comparison with historical orthophotos. The 39-year time series enables historical reconstruction of forest composition dynamics and long-term ecological change detection at national scale and relevant time scales for climate-driven species range shifts.

1 Introduction

Landsat for long term forest monitoring: With nearly 40 years of continuous acquisitions, the Landsat archive constitutes the longest consistent Earth observation record available for global vegetation monitoring (Wulder et al., 2022). Its four decades of continuity makes it uniquely suited for change detection or hindcasting of environmental variables including forest habitat characteristics (Bell et al., 2024; Turubanova et al., 2023). The archive has been applied to discriminate forest types and quantify forest variables such as canopy height or greening trends from regional to continental scales (Pflugmacher et al., 2019; Turubanova et al., 2023; **?**). In monitoring long-term changes and disturbances in forest vegetation under climate change, the Landsat afchive provides a valuable modelling base (Viana-Soto and Senf, 2025; Grünig et al., 2026). The relatively slow change and succession dynamics of forest vegetation, which is unfolding over decades, make this long temporal availability particularly valuable. A shorter observation period would fail to capture natural stand-level transitions such as species shifts or forest densification, which happen gradually over long time windows (Tong et al., 2023; Midolo et al., 2026). The medium spectral, spatial, and temporal resolution of Landsat (6 reflectance bands, 30 m ground sampling distance, and 8–16 day revisit time) is adequate for capturing annual phenological dynamics at stand level, which demonstrated to be critical for distinguishing tree species and functional leaf types (Kollert et al., 2021; Hemmerling et al., 2021; Wang et al., 2022).

Challenges: Despite these strengths, the use of multi-decadal Landsat data for vegetation mapping raises several methodological challenges. The number of usable cloud-free observations has increased since the early periods of the archive, due to the addition of new Landsat satellites. This systematic increase in observation density has been shown to cause spurious trends in annual maximum NDVI, where up to 50% of observed Landsat-based greening in alpine environments can be attributed to observational sampling bias rather than true vegetation change (Bayle et al., 2024). A second challenge arises from the failure of the Scan Line Corrector (SLC-off) on Landsat 7 ETM+, which introduces systematic spatial data gaps. Finally, spectral radiometric differences between sensors — arising from different sensors, band widths, and data processing procedures across Landsat missions — introduce inter-sensor discontinuities that may produce spurious changes in modelled variables across the 40-year record.

ML/DL for dominant tree species and leaf type cover mapping: Multitemporal satellite data consistently outperform single-date or mosaic-based approaches for tree species and leaf type mapping. Phenologically informative seasons — particularly leaf-off periods and autumn senescence — contribute disproportionately to species discrimination (Kollert et al., 2021; Tan et al., 2025). When sufficient temporal coverage is available, dense time series exploiting the full phenological cycle can outperform the inclusion of additional environmental predictor variables such as topography or climate indices (Hemmerling et al., 2021). These findings highlight temporal resolution and coverage as primary sources for improving species-level modelling and mapping performances.

Deep learning, and in particular Transformer-based architectures, provide a framework for processing raw, irregular, multi-band satellite time series (STS). Transformers process sequences of observations without requiring interpolation to a regular sequence, an essential property for long Landsat archives with varying observation density across decades and sensor platforms (Zerveas et al., 2020; Yuan et al., 2022). Self-supervised pretraining on large unlabelled STS or image time series datasets has been shown to improve representation quality and performance on downstream tasks, particularly in data-scarce settings (Yuan Yuan et al., 2020; Hiebl et al., 2025). Contrastive learning approaches exploit temporal and seasonal variability by training representations to be invariant across different observation windows of the same location — a property suited to stable land cover types such as forest stands (Mañas et al., 2021; Chen et al., 2020). Providing calibrated uncertainty estimates alongside predictions has emerged as an important requirement for ecological mapping applications, to enhance interpretability by users. Deep ensembles of independently trained networks deliver calibrated epistemic uncertainty that reliably identifies out-of-distribution inputs and unreliable predictions, when feature and label related domain shifts appear during the mapping phase (Sylvain et al., 2024; Hiebl et al., 2025).

Relevance for Italys forests: Italian forests are undergoing rapid and spatially complex changes. The Italian National Forest Inventory (INFC2015) documents a continuing expansion of forested area from approximately 10.46 million ha in INFC2005 to 10.98 million ha in INFC2015 driven by land abandonment and targeted reforestation programmes (Gasparini et al., 2022). At the same time, land abandonment in combination with climate change alters phenological characteristics and forest composition in favor of species with higher drought resilience (Herraiz et al., 2025; Chelli et al., 2017), while evergreen broad-leaved species are spreading in deciduous forest areas (Conedera et al., 2018). These processes are spatially heterogeneous, reflecting regional variety in land use changes, climate change, and historical species composition. Despite these dynamics, spatially explicit maps of forest composition and variables in Italy remain restricted to static maps with no continuous temporal resolution. The most recent national reference product, the Carta Forestale d'Italia 2020 (CFI2020), provides a single-epoch wall-to-wall forest map at 1:10,000 scale based on photo-interpretation of aerial orthophotos, but does not capture interannual dynamics or historical trajectories of change (Mattioli et al., 2025).

What will be done? Our study will fill this gap by providing data-driven, spatio-temporal continuous maps of leaf type cover (LTC) fractions and dominant tree species (DTS) probabilities for all forested areas in Italy - annually and at Landsat-native $30m$ resolution. We assemble and process 40-years of Landsat Collection-2 Level-2 time series from 1985 to 2025 (Landsat-4/5/7/8/9) and train multiple Time Series Transformer models (TSTpad) via contrastive pretraining and a fine-tuning workflow. The final product are annual $30m$ maps of i) 11 dominant tree species classes, and ii) LTC fractions for evergreen broad-leaved, deciduous broad-leaved and coniferous functional leaf type. Additional ensemble-based uncertainty estimations will be provided per-pixel and target variable.

2 Materials and Methods

2.1 Study Area

The study covers all forested areas of peninsular and insular Italy, spanning several biogeographical, climatic, and latitudinal regions and an altitudinal range from sea level to approximately $2500\,m.s.l.$[1] in the Alps. Italian forests fall within alpine, temperate, sub-Mediterranean, and Mediterranean climate zones (Chelli et al., 2017), and are dominated by four forest habitat types: broadleaved-deciduous and temperate coniferous forests concentrated in the Alps and Apennines, and broadleaved-evergreen and Mediterranean conif-

[1] needs citation

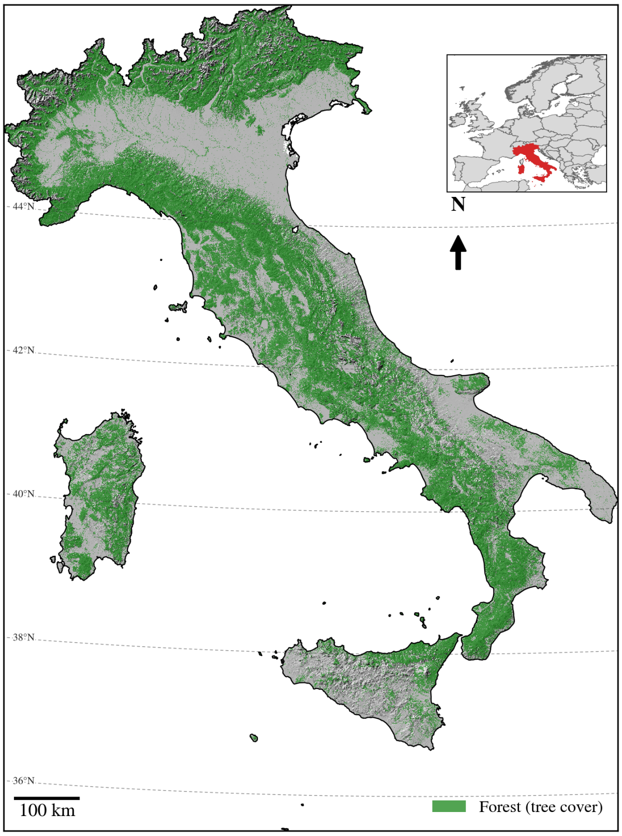

Figure 1. Study Area Italy with high forest cover in the Alps and mountainous areas of Central Italy.

erous forests concentrated along the coasts and in the south (Bricca et al., 2026). This climatic and ecological heterogeneity is a central challenge for wall-to-wall mapping, as models trained predominantly on well-sampled regions risk out-of-domain predictions in climatically and ecologically different areas with sparser reference data. Forests cover 37% of Italy's territory (10.98 million ha) (Gasparini et al., 2022). Italy's position as a transition zone between the Mediterranean and the temperate climate of Central Europe makes its forests especially vulnerable to climate change: warming and increasing drought frequency are already documented to shift phenology and favour more drought-resilient and evergreen broad-leaved species at the cost of deciduous ones (Chelli et al., 2017; Herraiz et al., 2025; Conedera et al., 2018). These climate-driven shifts compound ongoing changes in forest management and land use, jointly reshaping tree species distribution and biodiversity across the country (Bricca et al., 2026).

2.2 Modelling Data

figure: next to each other plot data images VPO, VDBI synthetic dataset and CFI polygonized data

2.2.1 Italian forest vegetation plot databases

The Forest Vegetation Database Italy (VDBI) is a compilation of over 16,000 phytosociological and vegetation relevé plot observations collected across Italy from the 1970s to the present. Each plot record contains species composition data, structural attributes, and cover estimates for the tree strata (Alessi et al., 2019). The VDBI covers the full range of Italian forest types from Mediterranean macchia to subalpine conifer belts. It provides temporal depth extending back to approximately 1972, enabling historical training data for pre-2000 Landsat time steps. The location accuracy of single plot observations in the dataset varies across observation year and source dataset. To avoid non-forest pixels leak into the training and test datasets, VDBI was cleaned manually based on orthophoto imagery. The final dataset after the cleaning process provides 9837 plot observations. Plot records include LTC estimates (evergreen broad-leaved, deciduous broad-leaved, coniferous) and tree species cover fractions, which were used to retrieve dominant DTS classification labels, allowing joint training of the regression $(VDBI_{DTS})$ and classification $(VDBI_{DTS})$ TSTpad model. A $10\times10\,km$ regular polygon grid was created for the train/validation/test split to reduce spatial autocorrelation between training and validation sets. A 20% validation split, stratified by polygon identifier is applied throughout all training stages (Safonova et al., 2023).

2.2.2 Field plot observations

The Vegetation Plot Observations (VPO) dataset consists of recent field campaigns conducted within Italian National Parks and protected areas. Each site comprises four systematic 100 m^2 vegetation plots arranged on a regular grid, within which evergreen broad-leaved cover is estimated for three vertical layers (0–1 m, 1–10 m, and above 10 m). The VPO data used in this study consists of observations from the 2024 field season, covering five National Parks (Sila, Gran Sasso, Gennargentu, Cilento, and Nebrodi) and yielding a total of 1440 individual plot observations. The VPO dataset provides high-quality, spatially clustered reference data for the most recent Landsat time steps (post-2020) and serves as the primary validation reference for model performance in the recent high-density observation period. The spatial clustering of VPO plots into national parks is explicitly accounted for in model evaluation by applying a cluster-based spatial cross-validation scheme, preventing over-optimistic accuracy estimates due to spatial autocorrelation (Safonova et al., 2023).

2.2.3 CFI Forest Inventory Data

The CFI2020 is the first national forest map of Italy produced at operational scale (1:10,000), based on photointerpretation of aerial orthophotos from 2018 to 2020 against three simultaneously applied forest definitions (Mattioli et al., 2025).

The CFI2020 delineates forested areas across all 21 Italian regions and provides stand-level attributes including forest category, silvicultural system, and dominant tree species group. In this study, the CFI2020 polygon layer serves two roles: i) as an additional training/validation data source for the DTS classification task, and ii) as a source of decision rules for cleaning the artificial LTC training dataset (see Section 2.1.4). In the first case two different datasets are created by i) randomly sampling a set of evenly distributed points over all classes, and ii) by sampling points from manually delineated pure stand polygons that are evenly distributed across classes. In the second case, CFI2020 species attribution is used to impose ecologically plausible constraints on predicted cover fractions of the synthetic dataset — for example, requiring that pixels mapped as Pinus-dominated stands contain at least 60% coniferous cover and fewer than 30% broad-leaved evergreen cover. This way we prevent ecologically non-feasible labels being forwarded into the training loop.

2.2.4 Synthetic leaf type cover data

Spatially explicit wall-to-wall leaf type cover data for model training are not available for Italy. Following the approach of Kang et al. (2021) and Tan et al. (2025) — who demonstrate the use of coarser or indirectly derived data sources as proxy training targets where direct measurements are absent — an artificial leaf type cover dataset was created from the VDBI and VPO plot observations. A Random Forest regression model was trained on the combined VDBI and VPO training split, with EVE, DEC, and CON cover fractions as prediction targets. As input features, AlphaEarth embeddings (Brown et al., 2025; Hiebl et al., 2026) — dense spectral-spatial-temporal representations derived from a globally pretrained geospatial foundation model — were extracted for all training plot locations. AlphaEarth embeddings were shown to achieve near-asymptotic species classification accuracy with very limited training data in Italian mountain forests (Ball et al., 2026), making them particularly suited as inputs for the pseudo-label generation step. Tessera embeddings (Feng et al., 2026) provide a fully open-source alternative with comparable performance characteristics.

The trained Random Forest model was then applied to the full CFI2020 forest extent to generate a nation-wide pixel-level pseudo-label layer for each cover fraction. This pseudo-label layer constitutes the primary continuous-cover target during the contrastive pretraining and early fine-tuning stages. To reduce label noise introduced by model prediction errors, the CFI2020 species attribution was used to apply ecologically motivated cleaning rules: pixels within pine-dominated forest polygons were required to have at least 60% coniferous cover and fewer than 30% broad-leaved evergreen cover; analogous rules were applied to pure-stand classes of beech, oak, and Mediterranean macchia. These cleaning rules remove the most egregious pseudo-label errors while preserving the spatial completeness of the training dataset.

2.2.5 Landsat time series

figure: landsat time series plot over all years for 2 different dominant tree species, include modelling approach with 3 year window aggregation

Annual Landsat time series were assembled for the full Italian national territory from the Google Earth Engine planetary-scale analysis environment using Landsat Collection 2 surface reflectance products from Landsat 4 TM (1982–1993), Landsat 5 TM (1984–2013), Landsat 7 ETM+ (1999–2023), Landsat 8 OLI/TIRS (2013–present), and Landsat 9 OLI-2/TIRS-2 (2021–present). Per-pixel cloud, cloud shadow, cirrus, and snow masking was performed using the Landsat Collection 2 QA PIXEL bitfield, decoded and applied via the ‘sattstools‘ preprocessing library (Hiebl, 2025a). Per-observation quality flags derived from QAPIXEL are propagated through the entire modelling pipeline as a key-padding mask, ensuring that cloud-contaminated or padded time steps are ignored by all attention layers of the TSTpad model. For each forested pixel, 14 spectral features are computed at each valid observation: six surface reflectance bands (blue, green, red, near-infrared, SWIR1, SWIR2) and eight derived spectral indices (NDVI, NIRv, NDMI, EVI, WDRVI, NBR, NDWI, GNDVI), computed using the ‘sattstools‘ index calculation routines (Hiebl, 2025a). The resulting multi-annual time series are stored as Zarr-compressed arrays to enable efficient random-access loading during model training.

2.2.6 Additional data sources

The ESA WorldCover product is used as an initial broad-scale delineation of forested areas across Italy. Only pixels labelled as tree cover are forwarded for prediction in the final maps. We are aware that the ESA WorldCover product does not provide a sufficient forest mask for detailed analysis of the datasets, as many mixed pixels and edge cases are remaining. But by applying a conservative masking during mapping we leave room for stricter forest delineation through other products without loosing regional accuracy.

2.3 Time Series Transformer Architecture

figure: model architecture and training scheme (including the contrastive pretraining and fine-tuning stages)

The backbone model is TSTpad, an encoder-only Time Series Transformer extended to support irregular, sparse, annual satellite image time series. The architecture employs multi-head self-attention across the temporal dimension, enabling the model to learn both short-range phenological features and long-range inter-seasonal dependencies within the aggregated 3-year observation window (Rußwurm and Körner, 2020; Zerveas et al., 2020).

For each target year, three consecutive years of Landsat observations are concatenated— encompassing the target year and the two immediately preceding years (year-2, year-1, year) — and aggregated into a single annual cycle to densify the sparse annual time series. After aggregation, observations are chronologically sorted and padded to a fixed sequence length of 100 time steps. Positions corresponding to cloud-masked or padded time steps receive a key-padding mask that prevents these positions from contributing to attention score computations (Vaswani et al., 2023). The actual acquisition day-of-year for each observation is encoded as a sinusoidal positional encoding, enabling the model to correctly represent the irregular temporal spacing of Landsat acquisitions and to distinguish phenological phases across the observation window without requiring interpolation to a regular grid (Yuan Yuan et al., 2020). Input features are normalised per band using a robust standardisation based on the median and interquartile range (quantiles q=0.1 and q=0.9) — with transform parameters fitted on the training data and frozen at inference time to prevent data leakage. A sensor platform embedding with $sensordim = 5$ encodes a multi-hot vector per year indicating which Landsat satellites (L4, L5, L7, L8, L9) contributed to the observations. This embedding is added to the pre-attention temporal encoding via a learned linear projection and makes the Transformer aware of inter-sensor radiometric differences over the 40-year archive. A post-attention pooling layer using a learned query vector collapses the temporal dimension into one single summary vector per output target, letting each targets learned query decide, which valid timesteps matter most for its prediction. In a setting with target classes from a wide range of phenological and spectral characteristics, this pooling strategy allows the model to learn class-specific temporal patterns across the observation window. The model heads are two-layer Multi-Layer-Perceptrons including Dropout and GeLU activation functions (Hendrycks and Gimpel, 2016). The classification model heads map the Transformer's output to class probabilities over eleven dominant tree species and three functional leaf type classes trained with two separate Cross Entropy Loss (CELoss) computations summed into a single loss value (dominant species and functional leaf type). With the dual loss the model learns to simultaneously capture species identity without losing the ability to capture the distinction between EVE and CON leaf types. The regression heads trained with Mean Squared Error Loss (MSELoss) maps the output to continuous EVE, DEC, and CON cover fractions. During contrastive

2.4 Training procedure

The model is trained via a three-stage progressive workflow designed to maximise temporal stability and to accommodate the heterogeneous quality and temporal coverage of available reference data.

Stage 1 — Contrastive pretraining. The backbone is trained using a joint loss that simultaneously optimises a supervised classification objective (DTS or) and an unsupervised NT-Xent contrastive loss (Chen et al., 2020): $L = \lambda \cdot$ NT-Xent + CE. For the contrastive term, positive pairs are formed from the same forest plot observed in two independently sampled three-year windows ($year_0$ and $year_1$ drawn at random), representing different but ecologically equivalent observation contexts for the same location. NT-Xent (normalised temperature-scaled cross-entropy; temperature $\tau = 0.1$) pushes the backbone representations of these positive pairs toward each other in embedding space while repelling representations of observations from different plots. This objective forces the backbone to learn year-invariant spectral-temporal features that are not confounded by observation density or inter-annual radiometric variability (Mañas et al., 2021). Positive pairs are further diversified by time series augmentation, which applies three independent stochastic perturbations to each view: random temporal masking (20% of valid time steps zeroed), spectral noise injection (additive Gaussian noise with $\sigma = 0.01$), and band masking (10% of spectral bands zeroed per sample) (Yuan et al., 2025). The contrastive and classification signals are routed through separate heads attached to the shared backbone: a two-layer MLP projection head (embedding dimension $\rightarrow$ 128) for NT-Xent, and the classification/regression head for CELoss or MSELoss, respectively. The model is optimised using AdamW with a cosine learning rate schedule preceded by a 10% warm-up phase; training runs for 60 epochs with a batch size of 128. Reference data consist of Italian forest vegetation plots from the VDBI and VPO datasets stored as Zarr-compressed arrays. Class imbalance is addressed jointly by class-weighted CrossEntropyLoss and by sample weights set to the square root of the inverse class frequency, following the approach of Grabska-Szwagrzyk et al. (2024). A stratified 20% validation hold-out, stratified by plot polygon identifier, is maintained across all training stages to prevent spatial autocorrelation between training and validation sets.

Stage 2 — Fine-tuning on labelled data. The contrastive backbone from Stage 1 is loaded and fine-tuned under a purely supervised objective (CrossEntropyLoss + MSELoss; NT-Xent dropped). The embedding layer is kept frozen in this stage to preserve the contrastive representations, while the task heads are retrained using a reduced learning rate to prevent catastrophic forgetting (Hiebl et al., 2025). This stage runs for 10 epochs on the full 1985–2025 Landsat time series, allowing the model to adapt the task heads to the full historical archive without disturbing the temporally invariant representations established during contrastive pretraining.

Stage 3 — Further fine-tuning on high-quality observations. Additional fine-tuning passes are applied from the Stage 2 checkpoint with progressively reduced and quality-filtered training subsets, giving priority to recent high-quality VPO field observations that provide the most accurate ground reference data for the present-day forest state. This stage is particularly intended to anchor the model's predictions of recent (post-2020) cover fractions to field-validated reference values, while the contrastive representations established in earlier stages ensure that this anchoring remains consistent with historical predictions.

Five independent model instances, differing in random seed (0–4), architecture micro-choices within TSTpad, and random subsets of training and validation data, are trained per stage. The five-seed ensemble delivers two outputs per pixel per year: the ensemble mean prediction (used as the primary data product) and the inter-seed standard deviation (used as the pixel-level uncertainty estimate). High inter-seed uncertainty identifies pixels where the model predictions are unstable, typically in regions of low observation density, unusual spectral composition, or high class confusion.

2.5 Mapping scheme

The forest prediction domain is defined by the intersection of the CFI2020 forest extent mask, the 5 m minimum canopy height layer, and the Copernicus HRL Forest Type product, with persistent snow-cover areas excluded. For each target year from 1985 to 2025, the corresponding three-year Landsat observation window (year-2 to year, inclusive) is assembled from the harmonised Landsat Collection 2 archive, cloud-masked, and processed to the 14 spectral features described in Section 2.1.5. Features are normalised with the fixed training-time 'TSRobustStandardize' parameters and padded to 100 time steps. The seven trained ensemble models are applied independently to each forested pixel, and the resulting class probability distributions and cover fraction predictions are aggregated across seeds to produce the mean prediction and inter-seed uncertainty layers.

Observation density varies substantially across the archive. Before 1999, when only Landsat 5 TM was operational, the three-year window typically yields fewer cloud-free observations than the post-2013 period when Landsat 8 and 9 operate simultaneously. This density gradient is an analogue of the observational sampling bias documented for NDVI trend analyses by Bayle et al. (2024): predictions in the early archive period carry higher uncertainty due to sparse temporal sampling, which may cause the model to have lower confidence in phenological-season attribution. This effect is explicitly quantified in the Technical Validation section (Section 4) and is flagged via the inter-seed uncertainty layer. Pixels with inter-seed uncertainty above a conservative threshold in the pre-1999 period should be interpreted with caution in trend analyses.

2.6 Evaluation

The validation strategy is designed to assess both the accuracy of the map product at individual time steps and the temporal consistency of predictions across the full 40-year archive. Three independent validation datasets are employed.

The VDBI hold-out split provides the primary historical validation reference, with plot observations dating back to approximately 1995. This split allows accuracy assessment at historical time steps (1995–2012) where field data are available, spanning the Landsat 5-only and Landsat 7/5 overlap periods. The VPO hold-out split — collected during the 2024 field season — provides the primary validation reference for recent predictions (post-2020) from the high-density Landsat 8/9 period. These two validation datasets provide complementary temporal coverage: the VDBI split quantifies historical accuracy under lower observation density, while the VPO split quantifies accuracy under the best available observation conditions.

A third validation source consists of a set of pure-stand forest polygons manually interpreted from high-resolution Google Earth Pro imagery at recent time steps. These polygons, selected to represent spectrally unambiguous mono-specific stands, enable validation of the classification output under conditions where the expected dominant species label is highly reliable, providing an upper bound on classification accuracy unconstrained by label uncertainty in mixed stands.

Accuracy metrics are reported separately for three observation-density periods: the historical Landsat 5-only period (pre-1999), the Landsat 7/8 transition period (1999–2013), and the recent high-density period (post-2013). For the classification head, overall accuracy (OA) and per-class F1 score are computed for each period. For the regression heads, root mean squared error (RMSE) and mean absolute error (MAE) are reported for each of the three cover fractions (EVE, DEC, CON).

To account for uncertainties in plot observations coming from (i) label noise, (ii) spatial mismatch between plot location and Landsat pixel, and (iii) temporal uncertainty and variations between plot observation year and Landsat acquisition year, additional sparse, probabilistic evaluation metrics are computed. Based on the plot location and reported year, a MonteCarlo approach is used to sample from the spatio-temporal uncertainty distribution and the best performance metrics uncluding uncertainty are reported for the sampling distribution. This approach reduces sensitivity to small misalignments and provides a more robust and uncertainty aware estimate of model performance at the stand level, while measuring the metric directly at the plot location and date might underestimate the model performance for each plot.

3 Conclusions

TEXT

4 Code and data availability

TEXT

Sample availability. TEXT

Appendix A

A1

Author contributions. TEXT

Competing interests. TEXT

Disclaimer. TEXT

Acknowledgements. TEXT

References

Alessi, N., Těšitel, J., Zerbe, S., Spada, F., Agrillo, E., and Wellstein, C.: Ancient refugia and present-day habitat suitability of native laurophylls in Italy, Journal of Vegetation Science, 30, 564–574, https://doi.org/10.1111/jvs.12743, 2019.

Ball, J. G., Wicklein, J. A., Feng, Z., Knezevic, J., Jaffer, S., Madhavapeddy, A., Atzberger, C., Dalponte, M., and Coomes, D.: Geospatial foundation models enable data-efficient tree species mapping in temperate mountain forests, https://doi.org/10.64898/2026.02.23.707022, iSSN: 2692-8205 Pages: 2026.02.23.707022 Section: New Results, 2026.

Bayle, A., Gascoin, S., Berner, L. T., and Choler, P.: Landsat-based greening trends in alpine ecosystems are inflated by multidecadal increases in summer observations, Ecography, 2024, e07 394, https://doi.org/10.1111/ecog.07394, _eprint: https://nsojournals.onlinelibrary.wiley.com/doi/pdf/10.1111/ecog.07394, 2024.

Bell, D. M., Gregory, M. J., and Yang, Z.: Hindcasting and updating Landsat-based forest structure mapping across years to support forest management and planning, Forest Ecology and Management, 572, 122 239, https://doi.org/10.1016/j.foreco.2024.122239, 2024.

Bricca, A., Zerbe, S., Sabatini, F. M., Hiebl, B., Rutzinger, M., di Musciano, M., Calvia, G., Chiarucci, A., Poschlod, P., Rossi, C., Alessi, N., and Bonari, G.: Topography and Soil Moisture Regulate the Temperature-Biodiversity Relationship of Forests, Global Ecology and Biogeography, 35, e70 186, https://doi.org/10.1111/geb.70186, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/geb.70186, 2026.

Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., Shelhamer, E., Lahera, E., Wiles, O., Ilyushchenko, S., Gorelick, N., Zhang, L. L., Alj, S., Schechter, E., Askay, S., Guinan, O., Moore, R., Boukouvalas, A., and Kohli, P.: AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data, https://doi.org/10.48550/arXiv.2507.22291, arXiv:2507.22291 [cs], 2025.

Chelli, S., Wellstein, C., Campetella, G., Canullo, R., Tonin, R., Zerbe, S., and Gerdol, R.: Climate change response of vegetation across climatic zones in Italy, Climate Research, 71, 249–262, https://doi.org/10.3354/cr01443, 2017.

Chen, T., Kornblith, S., Norouzi, M., and Hinton, G.: A Simple Framework for Contrastive Learning of Visual Representations, https://doi.org/10.48550/arXiv.2002.05709, arXiv:2002.05709 [cs], 2020.

Conedera, M., Wohlgemuth, T., Tanadini, M., and Pezzatti, G. B.: Drivers of broadleaved evergreen species spread into deciduous forests in the southern Swiss Alps, Reg Environ Change, 18, 425–436, https://doi.org/10.1007/s10113-017-1212-7, 2018.

Feng, Z., Atzberger, C., Jaffer, S., Knezevic, J., Sormunen, S., Young, R., Lisaius, M. C., Immitzer, M., Jackson, T., Ball, J., Coomes, D. A., Madhavapeddy, A., Blake, A., and Keshav, S.: TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis, https://doi.org/10.48550/arXiv.2506.20380, arXiv:2506.20380 [cs.LG], 2026.

Gasparini, P., Di Cosmo, L., Floris, A., and De Laurentis, D., eds.: Italian National Forest Inventory—Methods and Results of the Third Survey: Inventario Nazionale delle Foreste e dei Serbatoi Forestali di Carbonio—Metodi e Risultati della Terza Indagine, Springer Tracts in Civil Engineering, Springer International Publishing, Cham, ISBN 978-3-030-98677-3 978-3-030-98678-0, https://doi.org/10.1007/978-3-030-98678-0, 2022.

Grabska-Szwagrzyk, E., Tiede, D., Sudmanns, M., and Kozak, J.: Map of forest tree species for Poland based on Sentinel-2 data, Earth System Science Data, 16, 2877–2891, https://doi.org/10.5194/essd-16-2877-2024, 2024.

Grünig, M., Rammer, W., Senf, C., Albrich, K., André, F., Augustynczik, A. L. D., Baumann, M., Bohn, F. J., Bouwman, M., Bugmann, H., Collalti, A., Cristal, I., Dalmonech, D., De Coligny, F., Dobor, L., Dollinger, C., Espelta, J. M., Forrester, D. I., Garcia-Gonzalo, J., González-Olabarria, J. R., Hiltner, U., Hlásny, T., Honkaniemi, J., Huber, N., Jonard, M., Jönsson, A. M., Kunstler, G., Lagergren, F., Lindner, M., Mina, M., Moos, C., Morin, X., Muys, B., Nabuurs, G.-J., Nieberg, M., Patacca, M., Peltoniemi, M., Reyer, C. P. O., Schelhaas, M.-J., Storms, I., Thom, D., Toïgo, M., and Seidl, R.: Climate change will increase forest disturbances in Europe throughout the 21st century, Science, 391, eadx6329, https://doi.org/10.1126/science.adx6329, 2026.

Hemmerling, J., Pflugmacher, D., and Hostert, P.: Mapping temperate forest tree species using dense Sentinel-2 time series, Remote Sensing of Environment, 267, 112 743, https://doi.org/10.1016/j.rse.2021.112743, 2021.

Hendrycks, D. and Gimpel, K.: Gaussian Error Linear Units (GELUs), https://arxiv.org/abs/1606.08415v5, 2016.

Herraiz, A. D., Salazar-Zarzosa, P., Acosta-Muñoz, C., Hernández-Clemente, R., and Villar, R.: Aridity-induced phenological shifts and greening trends in Mediterranean forest species: Insights from 28 years of Landsat data in southern Spain, Ecological Indicators, 171, 113 115, https://doi.org/10.1016/j.ecolind.2025.113115, 2025.

Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Dorigo, W., Zerbe, S., and Rutzinger, M.: Advancing forest mapping: Pretraining strategies and deep-ensemble based uncertainty for predicting evergreen broad-leaved cover from Sentinel-2 time series, International Journal of Applied Earth Observation and Geoinformation, 142, 104 734, https://doi.org/10.1016/j.jag.2025.104734, 2025.

Hiebl, B., Alessi, N., Calvia, G., Bricca, A., Bonari, G., Zangari, G., Zerbe, S., and Rutzinger, M.: Combining specialized Sentinel-2 time series features with AlphaEarth Foundations for forest type mapping, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XI-3-2026, 117–124, https://doi.org/10.5194/isprs-annals-XI-3-2026-117-2026, conference Name: XXV ISPRS Congress 2026 "From Imagery to Understanding", Commission III - 4–11 July 2026, Toronto, Canada, 2026.

Kang, Y., Ozdogan, M., Gao, F., Anderson, M. C., White, W. A., Yang, Y., Yang, Y., and Erickson, T. A.: A data-driven approach to estimate leaf area index for Landsat images over the contiguous US, Remote Sensing of Environment, 258, 112 383, https://doi.org/10.1016/j.rse.2021.112383, 2021.

Kollert, A., Bremer, M., Löw, M., and Rutzinger, M.: Exploring the potential of land surface phenology and seasonal cloud free composites of one year of Sentinel-2 imagery for tree species mapping in a mountainous region, International Journal of Applied Earth Observation and Geoinformation, 94, 102 208, https://doi.org/10.1016/j.jag.2020.102208, 2021.

Mañas, O., Lacoste, A., Giró-i Nieto, X., Vazquez, D., and Rodríguez, P.: Seasonal Contrast: Unsupervised Pre-Training from Uncurated Remote Sensing Data, in: 2021 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 9394–9403, ISSN 2380-7504, https://doi.org/10.1109/ICCV48922.2021.00928, iSSN: 2380-7504, 2021.

Mattioli, W., R, R., D, B., G, C., G, D., D, G., M, P., and P, C.: La Carta Forestale d'Italia (CFI2020): un ritratto aggiornato dei boschi italiani, Forest@ - Journal of Silviculture and Forest Ecology, 22, 39, https://doi.org/10.3832/efor4836-022, 2025.

Midolo, G., Clark, A. T., Chytrý, M., Essl, F., Dullinger, S., Jandt, U., Bruelheide, H., Dengler, J., Axmanová, I., Aćić, S., Argagnon, O., Biurrun, I., Bonari, G., Chiarucci, A., Ćušterevska, R., De Frenne, P., De Sanctis, M., Divíšek, J., Doležal, J., Dziuba, T., Ejrnæs, R., Garbolino, E., Jentsch, A., Jiménez-Alfaro, B., Lenoir, J., Moeslund, J. E., Napoleone, F., Rumpf, S. B., Svenning, J.-C., Swacha, G., Tatarenko, I., Večeřa, M., Vynokurov, D., and Keil, P.: Sixty years of plant community change in Europe indicate a shift toward nutrient-richer and denser vegetation, Science Advances, 12, eaeb2493, https://doi.org/10.1126/sciadv.aeb2493, 2026.

Pflugmacher, D., Rabe, A., Peters, M., and Hostert, P.: Mapping pan-European land cover using Landsat spectral-temporal metrics and the European LUCAS survey, Remote Sensing of Environment, 221, 583–595, https://doi.org/10.1016/j.rse.2018.12.001, 2019.

Rußwurm, M. and Körner, M.: Self-attention for raw optical Satellite Time Series Classification, ISPRS Journal of Photogrammetry and Remote Sensing, 169, 421–435, https://doi.org/10.1016/j.isprsjprs.2020.06.006, 2020.

Safonova, A., Ghazaryan, G., Stiller, S., Main-Knorn, M., Nendel, C., and Ryo, M.: Ten deep learning techniques to address small data problems with remote sensing, International Journal of Applied Earth Observation and Geoinformation, 125, 103 569, https://doi.org/10.1016/j.jag.2023.103569, 2023.

Sylvain, J.-D., Drolet, G., Thiffault, É., and Anctil, F.: High-resolution mapping of tree species and associated uncertainty by combining aerial remote sensing data and convolutional neural networks ensemble, International Journal of Applied Earth Observation and Geoinformation, 131, 103 960, https://doi.org/10.1016/j.jag.2024.103960, 2024.

Tan, J., Li, J., Ma, T., Yan, X., and Huo, Z.: Leveraging Sentinel-1/2 time series and deep learning for accurate forest tree species mapping, Frontiers in Forests and Global Change, 8, https://doi.org/10.3389/ffgc.2025.1599510, 2025.

Tong, X., Brandt, M., Yue, Y., Zhang, X., Fensholt, R., Ciais, P., Wang, K., Liu, S., Zhang, W., Mao, C., and Jepsen, M. R.: Reforestation policies around 2000 in southern China led to forest densification and expansion in the 2010s, Communications Earth & Environment, 4, 260, https://doi.org/10.1038/s43247-023-00923-1, 2023.

Turubanova, S., Potapov, P., Hansen, M. C., Li, X., Tyukavina, A., Pickens, A. H., Hernandez-Serna, A., Arranz, A. P., Guerra-Hernandez, J., Senf, C., Häme, T., Valbuena, R., Eklundh, L., Brovkina, O., Navrátilová, B., Novotný, J., Harris, N., and Stolle, F.: Tree canopy extent and height change in Europe, 2001–2021, quantified using Landsat data archive, Remote Sensing of Environment, 298, 113 797, https://doi.org/10.1016/j.rse.2023.113797, 2023.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I.: Attention Is All You Need, https://doi.org/10.48550/arXiv.1706.03762, arXiv:1706.03762 [cs], 2023.

Viana-Soto, A. and Senf, C.: The European Forest Disturbance Atlas: a forest disturbance monitoring system using the Landsat archive, Earth System Science Data, 17, 2373–2404, https://doi.org/10.5194/essd-17-2373-2025, 2025.

Wang, M., Zheng, Y., Huang, C., Meng, R., Pang, Y., Jia, W., Zhou, J., Huang, Z., Fang, L., and Zhao, F.: Assessing Landsat-8 and Sentinel-2 spectral-temporal features for mapping tree species of northern plantation forests in Heilongjiang Province, China, Forest Ecosystems, 9, 100 032, https://doi.org/10.1016/j.fecs.2022.100032, 2022.

Wulder, M. A., Roy, D. P., Radeloff, V. C., Loveland, T. R., Anderson, M. C., Johnson, D. M., Healey, S., Zhu, Z., Scambos, T. A., Pahlevan, N., Hansen, M., Gorelick, N., Crawford, C. J., Masek, J. G., Hermosilla, T., White, J. C., Belward, A. S., Schaaf, C., Woodcock, C. E., Huntington, J. L., Lymburner, L., Hostert, P., Gao, F., Lyapustin, A., Pekel, J.-F., Strobl, P., and Cook, B. D.: Fifty years of Landsat science and impacts, Remote Sensing of Environment, 280, 113 195, https://doi.org/10.1016/j.rse.2022.113195, 2022.

Yuan, Y., Lin, L., Liu, Q., Hang, R., and Zhou, Z.-G.: SITS-Former: A pre-trained spatio-spectral-temporal representation model for Sentinel-2 time series classification, International Journal of Applied Earth Observation and Geoinformation, 106, 102 651, https://doi.org/10.1016/j.jag.2021.102651, 2022.

Yuan, Y., Lin, L., Xin, Q., Zhou, Z.-G., and Liu, Q.: An Empirical Study on Data Augmentation for Pixelwise Satellite Image Time-Series Classification and Cross-Year Adaptation, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 18, 5172–5188, https://doi.org/10.1109/JSTARS.2025.3527017, 2025.

Yuan Yuan, Yuan, Y., Lin, L., and Lei Lin: Self-Supervised Pretraining of Transformers for Satellite Image Time Series Classification, 14, 474–487, https://doi.org/10.36227/techrxiv.13025039.v2, 2020.

Zerveas, G., Jayaraman, S., Patel, D., Bhamidipaty, A., and Eickhoff, C.: A Transformer-based Framework for Multivariate Time Series Representation Learning, https://doi.org/10.48550/arXiv.2010.02803, 2020.